

Doohickey blueprint

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Optical elements

Item ID	Description	Designations
LB1757	F=30 mm, 25.4 mm biconvex lens	f_1
ME1-G01	25.4 mm first surface mirror	m
EBS1	25.4 mm 50:50 Beamsplitter	b
PL25mm	25mm Plossl eyepiece	f_2
LB—	F=20 mm, 12.4 mm biconvex lens	f_3

Compound lens formula:

$$f = \frac{1}{\frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}}$$

Image distance:

$$d_i = \frac{1}{\frac{1}{f} - \frac{1}{d_o}}$$

Object distance:

$$d_o = \frac{1}{2}d + \text{Offset}$$

Magnification:

$$M = -\frac{d_i}{d_o}$$

For the sake of simplicity the 25 mm telescope objective is considered to be a single 25 mm lens. The light coming from the collimator should be roughly parallel. For the eyepiece-mirror-inner lens system, the distance was measured at roughly 50.9075 mm. This gives an effective focal length as

$$f_{\text{IIT}} = \frac{1}{\frac{1}{f_1} + \frac{1}{f_2} - \frac{50.9075 \text{ mm}}{f_1 f_2}}$$

$$f_{\text{IIT}} = \frac{1}{\frac{1}{25 \text{ mm}} + \frac{1}{30 \text{ mm}} - \frac{50.9075 \text{ mm}}{25 \text{ mm} \times 30 \text{ mm}}}$$

$$f_{\text{IIT}} = 87.00636448 \text{ mm} \approx 87 \text{ mm}$$

My assumption is that the light from the thermal screen must be focused to the same spot from both the IIT and thermal screen. The light from the collimator can be considered to be an object at infinity, so $d_i = f_{\text{IIT}}$. As the lens is compound d_i will measure from the point in the center between the two lenses. This point will be

$$d_{i \text{ rel}} = 87 \text{ mm} - \frac{50.9075 \text{ mm}}{2} = 61.54625 \text{ mm}$$

From the center of the eyepiece; in this case it's also the center of our virtual lens. Based on the assumption above

$$d_{i \text{ IIT from eyepiece}} = d_{i \text{ Therm from eyepiece}}$$

Therefore we get

$$d_i = 61.546\,25\text{ mm} = \frac{1}{\frac{1}{f} - \frac{1}{\frac{1}{2}d+x}}$$

Where f is from the lens equation above.

$$\frac{1}{\frac{1}{f} - \frac{1}{\frac{1}{2}d+x}} = \frac{1}{\frac{1}{f_2} + \frac{1}{f_3} - \frac{d}{f_2 f_3} - \frac{1}{\frac{1}{2}d+x}}$$

From this point there's 3 unknowns: d , x and f_3 . Based on spacing constraints $d > 50\text{ mm}$ and f_3 must be a standard 12.2 mm lens from Thorlabs. This leaves x as the main unknown. Solving using sagemath we get:

$$x = -\frac{1}{2} \frac{2f_2 f_3 d_i + d_i d^2 + f_2 f_3 - (f_2 + f_3) d_i d}{f_2 f_3 - (f_2 + f_3) d_i + d_i d}$$

Plugging in values until something worked we get a value for $f_3 = 100\text{ mm}$, $d = 50$ and $x = 13.87\text{ mm}$